- THE VISIBLE UNIVERSE CONTAINS SOMEWHERE AROUND 1 SEPTILLION STARS:

  1 SEPTILLION = 1,000,000,000,000,000,000,000,000 = 10^{24}

  OF WHICH OUR STAR, THE SUN, IS SIMPLY ONE.

- WITH THAT IN MIND, IS THE SUN THE ONLY ONE OF THESE STARS THAT HAS AN ORBITING OBJECT WITH LIFE?

  OR, IN OTHER WORDS... ARE WE ALONE IN THE UNIVERSE?

- IF YOU THINK ABOUT IT, EITHER ANSWER TO THIS QUESTION IS TRULY PROFOUND...

YES: SEEMS TO BE WAY TOO ANTHROPOCENTRIC
"IT'S ALL ABOUT US". ANCIENT GREEKS?
WE ARE SPECIAL. COSMOLOGICAL PRINCIPLE?

NO: LIFE SHOULD BE COMMON
FERMI PARADOX - WHERE ARE THEY?

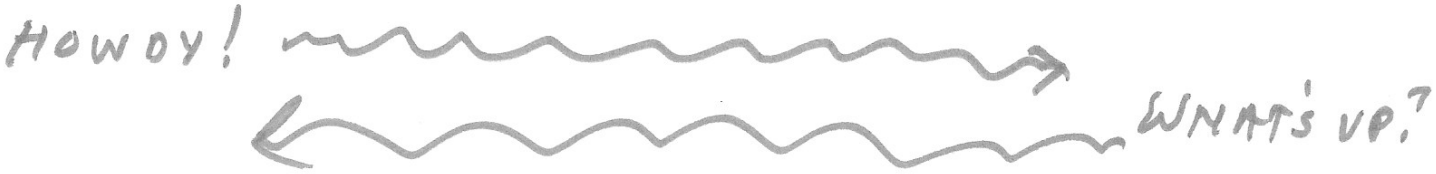
GET THE ANSWER? GO LOOK! WE CAN'T IT!
WE NEED TO LOOK FOR SIGNALS
→ LIMITED TO JUST "INTELLIGENT LIFE"

WHAT IS LIFE? REACT TO ITS ENVIRONMENT
TAKE IN NOURISHMENT
PRODUCE ENERGY
REPRODUCE

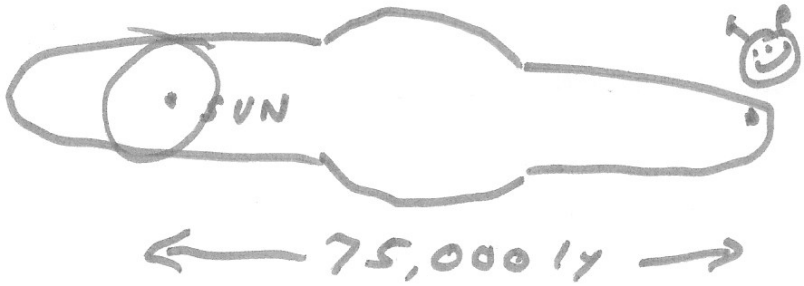
FIRE?

WHERE SHOULD WE LOOK?

TRAVEL? NOPE! SIGNALS? NOPE!



WE NEED ONLY CONSIDER THE M.W.
→ 200 BILLION STARS (STILL A LOT OF STARS!)



TRAVEL? NOPE! SIGNALS? NOPE!

WE REALLY ONLY NEED CONSIDER STARS ON OUR SIDE OF THE DISK (10 BILLION STARS).

SETI (GOVERNMENT FUNDED, ENDED IN A FEW YEARS)

PRIVATE SEARCHES -> PLANETARY SOCIETY

STILL TODAY, NO SIGNAL HAS BEEN FOUND...

WHAT TYPE OF SIGNAL?

- SPECTRUM? RADIO (DUST)
- WAVELENGTHS? WATER (H_2O)

  * H → 21 cm
  OH → 18 cm } WATER HOLE

  RADIO NOISE IS LOW

- POWER? OVER $\frac{1}{1,000}$ JANSKY
- CONTENT? MATH! ☺

SPECTRAL TYPES:

O B A F G K M

HOT ← COOL

BLUE WHITE RED

SHORT LIFE — LONG LIFE

O, B: EMIT TOO MUCH LETHAL UV LIGHT

O, B, A: TOO SHORT OF A LIFE

⋮

M: "GOLDILOCKS ZONE" IS VERY CLOSE TO THE STAR.

1. POWERFUL STELLAR FLARES EASILY DAMAGE A PLANET'S ATMOSPHERE.
2. PLANETS HERE WOULD BE TIDALLY LOCKED TO THE STAR, PERMANENT DAY/NIGHT SIDE.

WHAT'S LEFT? F, G, K → SUN-LIKE STARS

EVEN IF CONDITIONS ARE RIGHT FOR LIFE TO ARISE, DOES IT?

1950s: MILLER-UREY EXP.

PRIMORDIAL GASSES

→ CREATED AMINO ACIDS!

1960s : DRAKE EQUATION

$$N = r_* f_p n_E f_L f_I f_c L$$

N: NUMBER OF INTELLIGENT CIVILIZATIONS COMMUNICATING IN THE GALAXY TODAY.

r_*: RATE AT WHICH SUN-LIKE STARS FORM: (10/YEAR)

f_p: FRACTION OF THESE WITH PLANETS (1)

n_E: NUMBER OF EARTH-LIKE PLANETS ($\frac{1}{10}$)

f_L: FRACTION OF THESE WHERE LIFE ARISES (?)

f_I: FRACTION OF THESE WITH INTELLIGENT LIFE (?)

f_c: FRACTION OF THESE THAT DO COMMUNICATE (?)

L: LIFETIME OF THE CIVILIZATION (?)

PESSIMISTIC $N \approx 5$ (100,000 ly)

OPTIMISTIC $N = 1$ BILLION (35 ly)

SETI@HOME